Data Science Mids Guide

Easy notes from your lectures and handwritten notes, with Urdu examples, live graphs, scenario practice, quizzes and a Python code judge.

TOPIC 1 OF 9

Data Science & the lifecycle

What Data Science is, the steps of a project, and how to take a first look at data.

1 What is Data Science?

> 💡 **KEY POINT**
> Data Science means getting **useful insights from raw data**. It mixes **statistics**, **programming** and **domain knowledge**.

- Statistics: understand patterns and uncertainty.
- Programming: collect, clean and analyse data at scale (Python, R, SQL).
- Domain knowledge: know what the numbers mean for the real problem.

Exam definition: Data science is an interdisciplinary field that combines mathematics, statistics, computer science, artificial intelligence and domain expertise to extract meaningful insights and knowledge from raw, structured or unstructured data.

Examples: predicting student performance, detecting fraudulent bank transactions, recommending videos, predicting house prices.

اردو مثال

ڈیٹا سائنس کا مطلب خام ڈیٹا سے کارآمد معلومات نکالنا ہے۔ جیسے دکاندار پچھلے سالوں کی فروخت دیکھ کر اندازہ لگاتا ہے کہ عید پر کون سا سامان زیادہ بِکے گا۔ اس میں شماریات، پروگرامنگ اور اس شعبے کی سمجھ، تینوں شامل ہوتی ہیں۔

2 The Data Science lifecycle

> 💡 **KEY POINT**
> A lifecycle is a step-by-step process: **Understand → Acquire → Explore → Clean → Analyse → Engineer → Model → Evaluate → Deploy → Monitor**.

#	Stage	What you do
1	Problem / business understanding	Define the problem, objective, target variable, constraints and success criteria. (“Pehle samjho kya solve karna hai.”)
2	Data acquisition (collection)	Collect data from databases, APIs, files, sensors, websites, surveys.
3	Data understanding / exploration	Inspect structure, variables, distributions, missing values, duplicates.
4	Data cleaning / preparation	Fix missing values, duplicates, wrong formats, outliers, inconsistent data.
5	EDA and analysis	Use statistics and charts to discover patterns and relationships.
6	Feature engineering / selection	Create useful features and select relevant variables.
7	Model building (ML modelling)	Train machine-learning or statistical models.
8	Model evaluation	Test the model on **unseen rows** using suitable metrics.
9	Deployment	Put the model into practical use (make it live).
10	Monitoring and maintenance	Track performance, data drift, errors and when to retrain.

> ⚠️ **NOTE**
> Your handwritten notes list a shorter version (problem definition, data collection, data cleaning, EDA, ML modelling, evaluation, deployment and monitor). Frameworks may use different numbers or names of stages; the order of ideas is the same.

اردو مثال

فرض کریں آپ جاننا چاہتے ہیں کہ کون سے طلبہ فیل ہو سکتے ہیں۔ پہلے مسئلہ طے کریں، پھر حاضری اور نمبروں کا ڈیٹا جمع کریں، غلط یا خالی خانے ٹھیک کریں، چارٹ بنا کر پیٹرن دیکھیں، ماڈل بنائیں، نئے طلبہ پر اس کی جانچ کریں اور آخر میں اسے استعمال میں لا کر نگرانی کرتے رہیں۔

3 EDA and visualisation: the first look

KEY POINT

EDA (Exploratory Data Analysis) = explore the data to find patterns. **Data visualisation** = show data as charts and graphs.

From your notes, EDA always begins with: **shape** → **info** → **describe** → **null check**.

Step	Pandas call (illustrative)	Tells you
shape	`df.shape`	number of rows and columns
info	`df.info()`	column names, data types, non-null counts
describe	`df.describe()`	count, mean, std, min, quartiles, max
null check	`df.isnull().sum()`	missing values per column

```
import pandas as pd
df = pd.read_csv("students.csv")   # needs your own file
print(df.shape)
df.info()
print(df.describe())
print(df.isnull().sum())
```

This code needs a CSV file named students.csv; it is shown for learning and was not run here.

📝 Mid-exam style questions (Topic 1)

Define Data Science and explain the lifecycle.

Why is evaluation done on unseen data?

Why is monitoring needed after deployment?

TOPIC 2 OF 9

Roles in Data Science

Who does what in a data team, and which tools each role uses.

1 The main roles and their tools

KEY POINT

Data Scientist = **models and statistics**. Data Analyst = **reports and visualisation**. Data Engineer = **pipelines**. ML Engineer = **deploys models**.

Role	Main job	Tools (from your notes)
Data Scientist	Modelling and statistics; analyses data and builds predictive models	Python, R, SQL, TensorFlow
Data Analyst	Reporting and visualisation; dashboards	Excel, SQL, Tableau, Power BI
Data Engineer	Builds data pipelines and infrastructure	Spark, Hadoop, Airflow, Kafka
ML Engineer	Deploys models to production and maintains ML systems	PyTorch, TensorFlow, Docker
Data Architect	Designs data architecture and integration	Databases, cloud, architecture
Business / Domain Analyst	Connects business needs with data solutions	Domain knowledge, communication

اردو مثال

ہسپتال میں: ڈیٹا انجینئر مریضوں کے ریکارڈ کی پائپ لائن بناتا ہے، ڈیٹا اینالسٹ ماہانہ رپورٹ بناتا ہے، ڈیٹا سائنٹسٹ بیماری کی پیشگوئی کا ماڈل بناتا ہے اور ایم ایل انجینئر اسے اصل نظام میں چلاتا ہے۔

2 Telling the roles apart (exam focus)

> 💡 **KEY POINT**
> The **value** of all these roles is **business insight** from data.

- **Data Engineer** builds infrastructure → data flows reliably.
- **Data Analyst** describes what happened (reporting, descriptive analysis).
- **Data Scientist** builds models and finds insight, often predicting what will happen.
- **ML Engineer** turns a model into a working, monitored product.

Common mix-up: calling every data job "data scientist". In real teams the roles overlap, but each has a main responsibility.

Practise: try the scenarios "Who does which job?" and "The startup's first data hire".

📝 Mid-exam style questions (Topic 2)

Explain the roles of Data Scientist, Data Analyst, Data Engineer and ML Engineer.

Which role builds infrastructure and which builds models?

TOPIC 3 OF 9

Big Data (the 5 Vs)

What makes data “big” and the five Vs that describe it.

1 What is Big Data?

💡 KEY POINT

Big Data = very large, complex, fast-growing or diverse datasets that traditional processing methods may struggle to handle.

It is not only about size. Data can also be big because it arrives very fast, comes in many forms or is hard to trust.

اردو مثال

واٹس ایپ پر ہر سیکنڈ لاکھوں پیغامات (حجم)، بہت تیزی سے (رفتار)، متن، تصویر، آواز اور ویڈیو (اقسام)، کچھ جعلی اکاؤنٹس (سچائی) اور اس سے اشتہارات کا فائدہ (قدر)۔

2 The 5 Vs

V	Question it answers	Meaning	Example
Volume	How much?	Amount of data	Terabytes, petabytes
Velocity	How fast?	Speed at which data is generated and processed	Real-time data, stock ticks
Variety	How many forms?	Different types: structured, semi-structured, unstructured	Tables, JSON, images, audio
Veracity	How trustworthy?	Accuracy and reliability of the data	Fake accounts, wrong records
Value	How useful?	Business insight obtained from data	Higher sales, fewer frauds

> 💡 **KEY POINT**
> Memory trick: **much – fast – forms – trust – use**.

> ⚠️ **NOTE**
> In the notes' last line the term "accuracy and trustworthiness" belongs to **Veracity** (not Velocity).

📝 Mid-exam style questions (Topic 3)

What is Big Data? Explain the 5 Vs with examples.

TOPIC 4 OF 9

Data types

How data is organised (structure) and what kind each variable is.

1 Structured, semi-structured and unstructured data

	Structured	Semi-structured	Unstructured
Shape	Fixed rows and columns	Flexible, uses keys / tags	No predefined table
Examples	SQL tables, bank or student database	JSON, XML	Images, text, audio, video
Typical tools	SQL, spreadsheets	JSON/XML parsers, APIs	Image / text / audio processing

Example of unstructured data use: digital image processing, like the filters in Snapchat.

اردو مثال

اسکول کا داخلہ رجسٹر (خانوں والا) ساختہ ڈیٹا ہے، کسی ایپ کی JSON فائل نیم ساختہ ہے اور موبائل میں رکھی تصویریں اور وائس نوٹس غیر ساختہ ڈیٹا ہیں۔

2 Types of variables

Type	Meaning	Example
Numerical: discrete	Counted whole numbers	Number of students
Numerical: continuous	Measured on a scale	Height, temperature
Categorical: nominal	Labels with **no** order	City, product category
Categorical: ordinal	Labels with a **meaningful order**	Low / medium / high
Boolean	True/False or 1/0	Owns a laptop
Text	Natural-language strings	Reviews
Date/time	Dates, timestamps, durations	Order time

> 💡 **KEY POINT**
> Exam focus: nominal has **no** natural order; ordinal **has** a meaningful order.

📝 Mid-exam style questions (Topic 4)

Differentiate structured, semi-structured and unstructured data.

Nominal vs ordinal data?

TOPIC 5 OF 9

Data acquisition

Where data comes from and how to judge whether it is good enough.

1 What is data acquisition?

> 💡 **KEY POINT**
> Data acquisition = **obtaining data** from relevant sources for analysis. Data itself is **raw facts and figures**.

Forms of data (from your notes): **pictorial** (images), **textual** (words) and **digital** (stored electronically).

Sources: databases, APIs, CSV / Excel / JSON files, websites, sensors, surveys, logs and cloud systems.

2 Primary and secondary data

	Primary data	Secondary data
Meaning	Collected **directly** for the current purpose	Already collected by someone else
Examples (notes)	Interview, survey, questionnaire (also experiments, sensors)	Repository, database data, cloud (also government and research datasets)
Good	Fits your question exactly	Quick and cheap
Watch out	Costly and slow	May not fit your question; check quality and permission

اردو مثال

آپ خود طلبہ سے سوالنامہ بھروائیں تو یہ بنیادی (primary) ڈیٹا ہے۔ اگر آپ حکومت کی شائع کردہ مردم شماری کا ڈیٹا ڈاؤن لوڈ کریں تو یہ ثانوی (secondary) ہے۔

3 Concerns when acquiring data

- **Permission:** do you have the right to use it (terms of use, consent)?
- **Privacy:** protect personal details.
- **Quality:** accurate, complete, consistent?
- **Relevance and reliability:** does it answer your question, and can you trust the source?
- **Cost and format:** can you afford and read it?
- **Collection bias:** does the sample represent everyone?

4 Data quality dimensions

Dimension	Question	Example problem
Accuracy	Is it correct?	Age = 250
Completeness	Are values present?	Missing marks
Consistency	Same format everywhere?	“Lahore” vs “lahore”
Validity	Does it follow the rules?	Negative quantity
Uniqueness	No unwanted duplicates?	Same student twice
Timeliness	Is it up to date?	Last year's prices

📝 Mid-exam style questions (Topic 5)

What is data acquisition? Explain primary and secondary data.

TOPIC 6 OF 9

Statistics for Data Science

Describing data, the bell curve, hypothesis testing and correlation.

1 Descriptive statistics

KEY POINT

Statistics helps with **data collection** and **distribution**; Data Science uses it for **data analysis** and to **understand data patterns**.

Concept	Formula	Explanation	Example
Mean	sum / count	Average value	Average of marks
Median	middle value (sorted)	Value in the middle	Salary with outliers
Mode	most frequent value	Occurs most often	Fashion / surveys
Variance	average of squared deviations	How spread out the data is	Stock volatility
Std deviation	√variance	Average distance from the mean	Test-score spread
Correlation	r = Cov(X,Y) / (sx · sy)	Relationship between two variables	Height vs weight
p-value	probability under H0	< 0.05 → result significant	Drug trial result

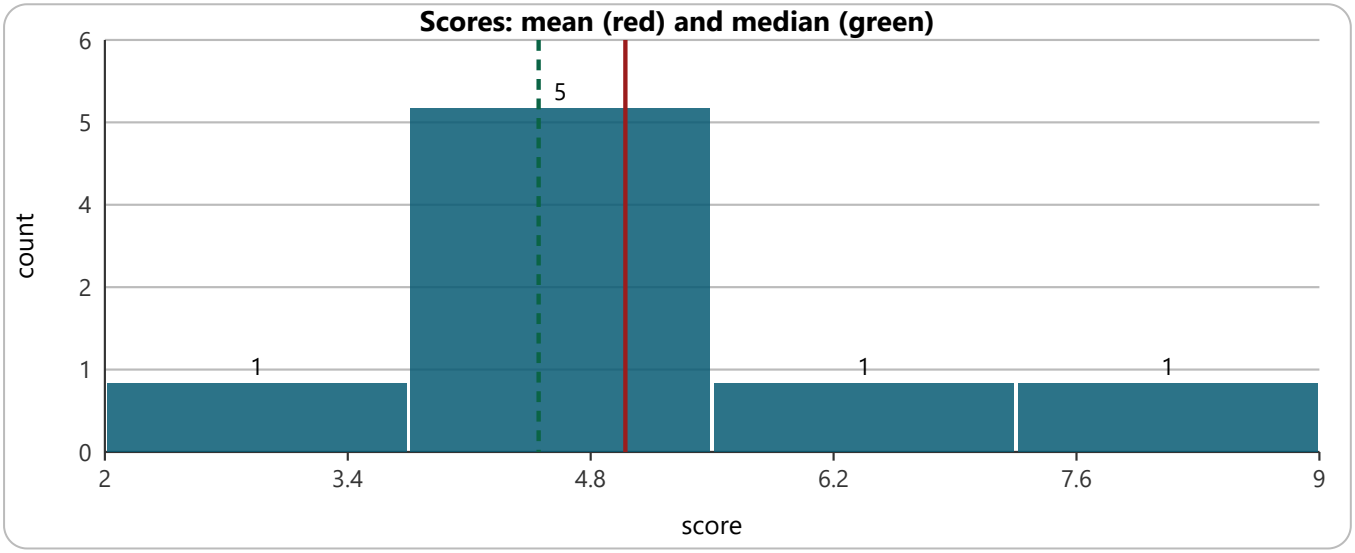
اردو مثال

پانچ ملازمین کی تنخواہ 30، 32، 35، 36 ہزار اور مالک کی 400 ہزار ہو تو اوسط (mean) بہت بڑھ جاتا ہے مگر درمیانی قدر (median) تقریباً ٹھیک رہتی ہے۔ اس لیے تنخواہ کے لیے median بہتر ہے۔

2 Spread, range and outliers

Worked example with the scores 2, 4, 4, 4, 5, 5, 7, 9:

1. Mean = (2+4+4+4+5+5+7+9) / 8 = 40 / 8 = **5**
2. Squared deviations: 9, 1, 1, 1, 0, 0, 4, 16 → sum = 32
3. Variance = 32 / 8 = **4**
4. Standard deviation = √4 = **2**
5. Range = 9 − 2 = **7**



KEY POINT

The **mean is sensitive** to extreme values; the **median is robust**. An **outlier** is an unusual value: investigate it, do not delete it blindly.

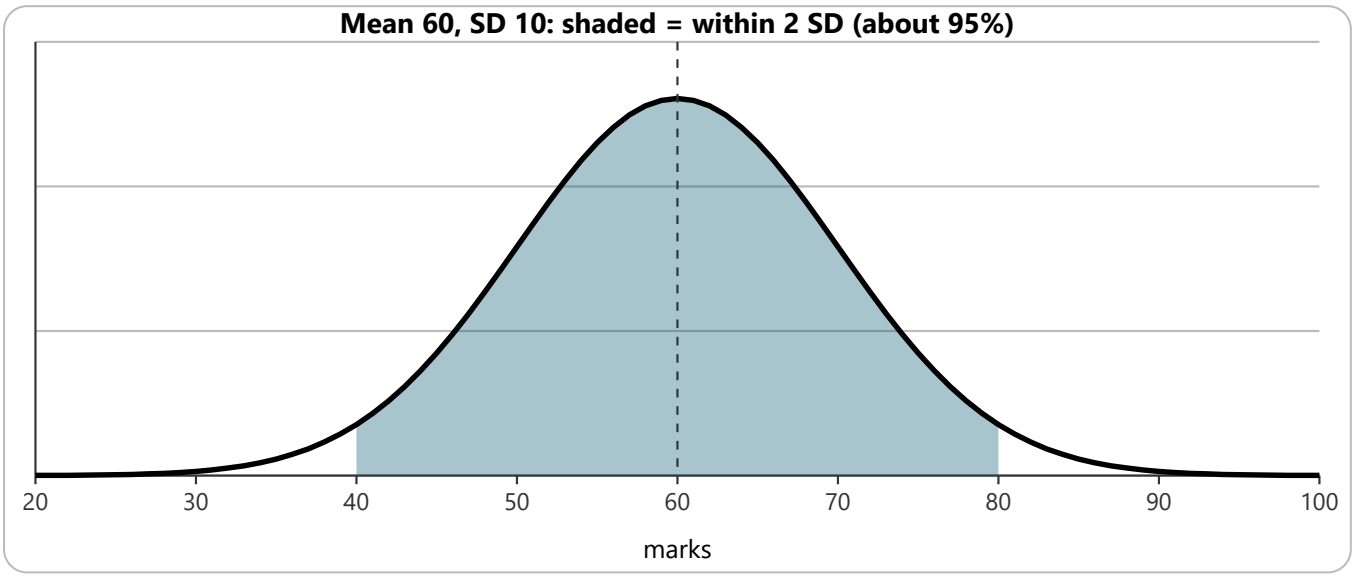
NOTE

Population variance divides by **n** (as in the lecture definition); when your data is only a sample, software often divides by **n − 1**.

3 Normal distribution and z-score

KEY POINT

The **normal distribution** is the **bell curve**: **68%** of data lies within 1 standard deviation of the mean, **95%** within 2 and **99.7%** within 3.



z-score = (x − mean) / SD: how many standard deviations a value is from the mean. A mark of 80 with mean 60 and SD 10 has z = 2.

اردو مثال

کلاس کے نمبر عموماً گھنٹی کی شکل میں ہوتے ہیں: زیادہ تر طلبہ اوسط کے قریب اور بہت کم طلبہ بہت اوپر یا بہت نیچے۔

4 Hypothesis testing and the p-value

> 💡 **KEY POINT**
> **Steps:** define H0 → define H1 → set alpha (0.05) → run the test → get the p-value → compare p with alpha → decision.

- **H0 (null):** the default claim: no difference, no effect.
- **H1 (alternative):** what you suspect instead.
- **Alpha (significance level):** the cut-off, usually 0.05.
- **p < alpha** → reject H0 (result significant). **p ≥ alpha** → fail to reject H0.

> ⚠️ **NOTE**
> **p-value** = the probability of a result at least this extreme **assuming H0 is true**. It is **not** the probability that H0 is true, and "fail to reject" does not prove H0.

اردو مثال

ایک کمپنی کہتی ہے کہ اس کی ڈیلیوری اوسطاً 30 منٹ میں ہوتی ہے۔ ہم نمونہ لے کر دیکھتے ہیں؛ اگر p-value 0.05 سے کم آئے تو دعوے کے خلاف مضبوط ثبوت ہے۔

5 Correlation vs causation

> 💡 **KEY POINT**
> Correlation r runs from **−1** (perfect negative) through **0** (none) to **+1** (perfect positive). It shows **association**, not cause.

|r|	Strength
0.8 – 1.0	very strong
0.6 – 0.8	strong
0.4 – 0.6	moderate
0.2 – 0.4	weak
below 0.2	very weak / none

Notes example: ice-cream sales and drowning both rise in hot weather: correlation exists, causation does not. Hot weather (a **confounder**) drives both.

اردو مثال

گرمی میں آئس کریم بھی زیادہ بکتی ہے اور لوگ تیراکی بھی زیادہ کرتے ہیں، اس لیے ڈوبنے کے واقعات بھی بڑھتے ہیں۔ آئس کریم اور ڈوبنا ساتھ ساتھ بڑھتے ہیں (correlation) مگر ایک دوسرے کی وجہ نہیں (causation نہیں)۔

6 Probability and sampling basics

- **Probability** measures how likely an event is, from 0 (impossible) to 1 (certain): P = favourable outcomes / all outcomes. Example: a fair die shows 4 with probability 1/6.
- **Population** = everyone you want to know about. **Sample** = the subset you actually measure.
- Common sampling methods: **simple random** (everyone has an equal chance), **systematic** (every k-th item), **stratified** (random from each group), **cluster** (randomly choose whole groups).
- A biased sample (for example only one city) gives misleading results.

📝 Mid-exam style questions (Topic 6)

Explain mean, median, mode, variance and standard deviation.

State the steps of hypothesis testing.

Correlation vs causation with an example.

TOPIC 7 OF 9

Classification metrics

How to check how well a classification model works: confusion matrix, accuracy, precision, recall and F1.

1 What is classification?

KEY POINT

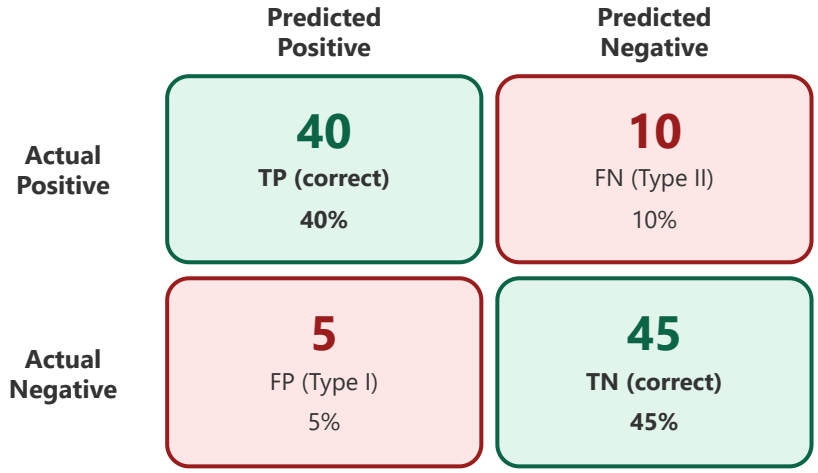
Classification is a **supervised** machine-learning task that predicts a **categorical class or label**.

- **Binary:** two classes (spam / not spam).
- **Multiclass:** more than two (cat / dog / bird).
- **Multilabel:** one item can have several labels.

Examples: fraud / not fraud, positive / negative review, approved / rejected. The notes say metrics help us **check how well the model works**.

2 The confusion matrix: TP, TN, FP, FN

> 💡 **KEY POINT**
> A confusion matrix compares **actual** classes with **predicted** classes. Read each term as: **(was the model right?) + (what did it predict?)**.



Term	Meaning (fraud example from your notes)
TP true positive	Model said “fraud” and it really was fraud ✓
TN true negative	Model said “no fraud” and it really was genuine ✓
FP false positive	Model said “fraud” but it was genuine. Also called **Type I error** (false alarm).
FN false negative	Model said “no fraud” but it was fraud. Also called **Type II error** (missed case).

We use **TP = 40, TN = 45, FP = 5, FN = 10** (total 100) in all worked examples below.

اردو مثال

فراڈ والا ماڈل: اگر اس نے اصلی فراڈ کو 'فراڈ' کہا تو یہ TP ہے۔ اگر اصلی لین دین کو 'فراڈ' کہہ دیا تو FP (جھوٹا الارم) ہے۔ اگر اصلی فراڈ کو 'ٹھیک' کہہ دیا تو FN (چھوٹ جانے والا کیس) ہے۔

3 Accuracy

> 💡 **KEY POINT**
> Accuracy = **(TP + TN) / (TP + TN + FP + FN)**: how many predictions were right in total (right / total).

Worked: (40 + 45) / (40 + 45 + 5 + 10) = 85 / 100 = **0.85** (85%).

> ⚠️ **NOTE**
> **Careful:** on **imbalanced** data accuracy can mislead. If only 10 of 1,000 transactions are fraud, a model that always says "no fraud" is 99% accurate and catches nothing.

4 Precision

> 💡 **KEY POINT**
> Precision = **TP / (TP + FP)**: of all the "yes" predictions, how many were actually right.

Worked: 40 / (40 + 5) = 40 / 45 = **0.889**.

Use it when: a **false alarm is costly**, like a genuine email being marked as spam. Goal: reduce false alarms.

5 Recall and specificity

> 💡 **KEY POINT**
> Recall (sensitivity) = **TP / (TP + FN)**: of all the actual "yes" cases, how many did the model catch.

Worked: 40 / (40 + 10) = 40 / 50 = **0.80**.

Use it when: missing a real case is dangerous (disease, fraud). Goal: reduce missed cases.

Specificity = TN / (TN + FP) = 45 / 50 = **0.90**: of all actual "no" cases, how many were correctly found.

> اردو مثال
>
> اسپیم فلٹر میں اصلی ای میل کا اسپیم میں چلا جانا مہنگا ہے، اس لیے precision اہم ہے۔ بیماری کی اسکریننگ میں بیمار کا چھوٹ جانا خطرناک ہے، اس لیے recall اہم ہے۔

6 F1 score

> 💡 **KEY POINT**
> F1 = **2 × (Precision × Recall) / (Precision + Recall)**: the **harmonic mean**, a balance of precision and recall.

Worked: 2 × (0.889 × 0.80) / (0.889 + 0.80) = 1.422 / 1.689 = **0.842**.

Use it when both precision and recall matter. It is low if either one is low. (Notes: "jab precision dono important ho".)

7 Which metric should I use?

Situation	Costly error	Focus on
Spam filter (genuine email marked spam)	False Positive	**Precision**
Disease screening, fraud detection	False Negative	**Recall**
Both matter, or imbalanced classes	Both	**F1** (with precision and recall)
Balanced classes, quick overview	—	Accuracy

> 💡 **KEY POINT**
> Exam focus: do **not** automatically choose accuracy. In disease screening a false negative can be especially serious.

📝 Mid-exam style questions (Topic 7)

Draw and explain a confusion matrix. Define TP, TN, FP, FN.

Write the formulas for accuracy, precision, recall, specificity and F1.

Why is accuracy misleading on imbalanced data?

TOPIC 8 OF 9

Overfitting & underfitting

Why a model can fail on new data, and how to tell what went wrong.

1 The big idea: generalisation

> 💡 **KEY POINT**
> A model learns from **training data**, but its real job is to perform well on **new data it has never seen**. This ability is called **generalisation**. Overfitting and underfitting are two ways generalisation fails.

2 Underfitting

> 💡 **KEY POINT**
> The model is **too simple**. It has not learned the pattern, so it performs **poorly on both** training data and new data.

Also called **high bias**. Causes: too few features, short training, strong regularisation, a model too simple for the pattern.

3 Overfitting

> 💡 **KEY POINT**
> The model is **too complex**. It **memorises the training data including its noise**, so it looks excellent in training but performs poorly on new data.

Also called **high variance**. Causes: too complex a model, too many features, a small dataset, excessive training.

4 Good fit and the student analogy

> 💡 **KEY POINT**
> **Good fit:** the model captures the true patterns and performs well on **both** training data and new data.

- **Underfitting** = a student who barely studied.
- **Overfitting** = a student who memorised last year's paper word for word.
- **Good fit** = a student who understands the concepts and can solve any new question.

اردو مثال

ایک طالب علم نے پچھلے سال کا پرچہ رٹ لیا (overfitting): پرانے سوال ٹھیک مگر نیا سوال غلط۔
دوسرے نے بالکل پڑھا ہی نہیں (underfitting): ہر سوال غلط۔ تیسرے نے تصورات سمجھے (good fit): نیا سوال بھی حل کر لیتا ہے۔

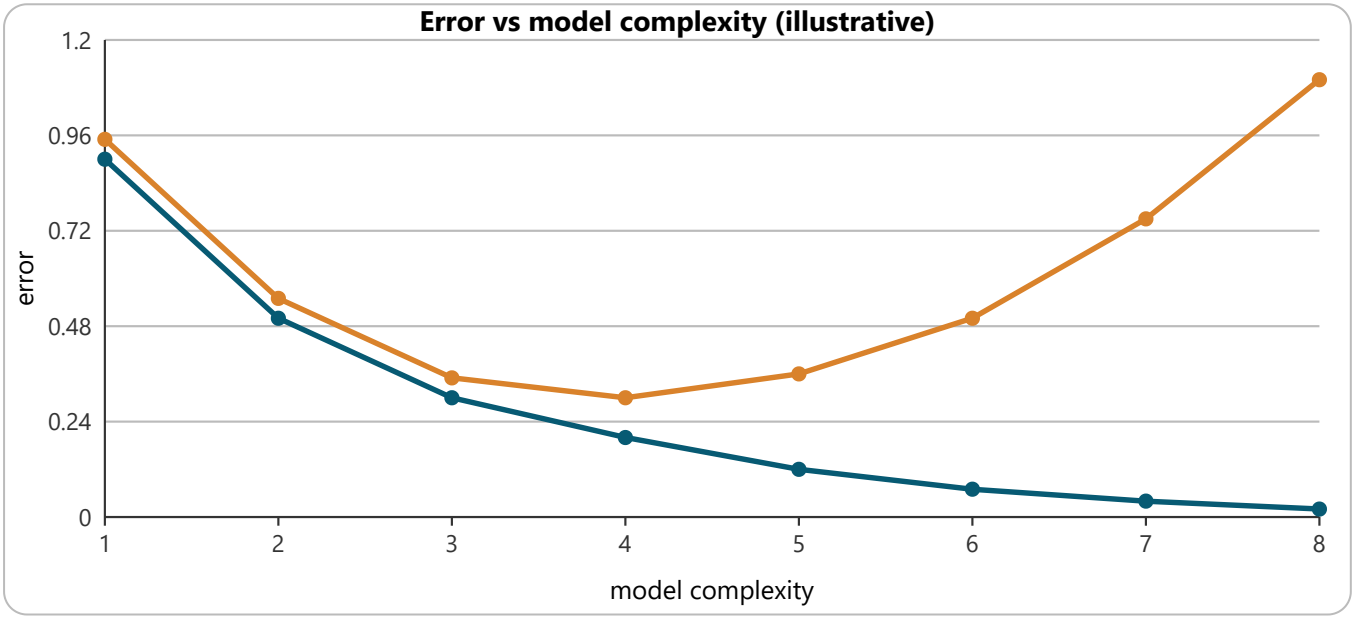
5 Underfitting vs overfitting: comparison table

Aspect	Underfitting	Overfitting
Meaning	Model too simple to learn the pattern	Model learns pattern plus noise
Also called	High bias	High variance
Training error	High	Very low
Test error	High	High
Gap between train and test	Small	Large
Model complexity	Too low	Too high
Typical cause	Few features, short training, strong regularisation	Small data, too many features
Data needed	More features or a better model may help	More data usually helps
Main fix	Increase complexity, train longer	Simplify, regularise, add data
Learning curve	Both curves high and close together	Training low, validation high, wide gap
Analogy	Student who did not study	Student who memorised answers

6 Quick diagnosis with numbers

Training accuracy	Test accuracy	Diagnosis
58%	55%	**Underfitting**: both are low
98%	70%	**Overfitting**: large gap
90%	88%	**Good fit**: both high and close

Test error is lowest at a **medium** complexity. Beyond that the model starts to overfit:



> ⚠ **NOTE**
> The numbers in the table are illustrative, as in your notes. In practice, judge by both the level and the gap, not by a fixed cut-off.

7 How to fix them

Problem	Solutions
Underfitting	Use a model with suitable complexity, add useful features, reduce excessive regularisation, train appropriately.
Overfitting	Simpler model, regularisation, cross-validation, feature selection, pruning, early stopping, more representative data.

Rule of thumb: **both errors high → add capacity; big gap → reduce complexity or add data**.

📝 Mid-exam style questions (Topic 8)

What is overfitting? Give causes and solutions.

What is underfitting? Give causes and solutions.

Compare training and test error for underfitting, good fit and overfitting.

TOPIC 9 OF 9

Tools, cleaning & charts

Python libraries, three tools (Python, Orange, Weka), data cleaning, manipulation and visualisation.

1 Python libraries for Data Science

Library	Purpose	Key functions (from your notes)
NumPy	Numerical computing, arrays	`np.array(), np.mean(), np.dot()`
Pandas	Data manipulation, DataFrames	`pd.read_csv(), df.groupby(), df.merge()`
Matplotlib	Basic plotting	`plt.plot(), plt.bar(), plt.scatter(), plt.show()`
Seaborn	Statistical visualisation	`sns.heatmap(), sns.boxplot(), sns.pairplot()`
Scikit-learn	Machine learning	`train_test_split()`
TensorFlow / Keras	Deep learning	`model.compile(), model.fit()`
SciPy	Scientific and statistical computing	tests and optimisation
Statsmodels	Statistical modelling	regression and inference

> 💡 **KEY POINT**
> NumPy = numerical · Pandas = data · Matplotlib / Seaborn = visualisation · Scikit-learn = ML. Install with `pip install numpy`.

```
import numpy as np
a = np.array([1, 2, 3, 4])
print(a.mean())      # 2.5
print(np.dot(a, a))  # 30
```

This snippet needs NumPy installed; the outputs shown are what Python prints (checked).

2 Python vs Orange vs Weka

Tool	What it is	Think of it as	Best for
Python	A programming language; write code with libraries like pandas, seaborn and scikit-learn	A full workshop: you can build anything	Real projects; can be automated and wrapped in an API (FastAPI / Flask) or web app (Streamlit)
Orange	A visual tool: drag boxes (widgets) and connect them with lines	LEGO blocks: easy to build, easy to see	Demos and teaching; **not** designed for production systems
Weka	A Java tool with menus and tabs (Explorer)	A ready-made lab machine: press the right buttons, get results fast	Classic machine learning

Saving work: Python: save charts as images, write a PDF report, save the model with joblib. Weka: save the model as a `.model` file and reuse it in Weka or Java. Orange: save the workflow as an `.ows` file and export charts as images.

3 Data cleaning

KEY POINT

Data cleaning = detecting and correcting **inaccurate, incomplete, inconsistent, duplicate or irrelevant** data. Notes: fix weak or missing data; incorrect and duplicate values.

Problem	What to do
Missing values	Remove, or **impute** (mean / median / mode) when justified
Duplicates	Identify and remove repeated records
Incorrect data types	Convert to the right type (text → number, text → date)
Inconsistent values	Standardise spelling, case, units, categories
Outliers	Detect, then **investigate**: error or real value?
Invalid values	Check ranges and rules (impossible age)
Integration issues	Resolve differences when combining sources

KEY POINT

Exam focus: cleaning is **not** simply deleting bad rows. First understand *why* a value is missing, wrong or unusual.

اردو مثال

رجسٹر میں ایک طالب علم کا نام دو بار لکھا ہے (ڈپلیکیٹ)، کسی کے نمبر خالی ہیں، 'Lahore' اور 'lahore' الگ لکھے ہیں اور کسی کی عمر 250 لکھی ہے۔ صفائی میں ان سب کو دیکھ کر ٹھیک کیا جاتا ہے۔

4 Data manipulation

> 💡 **KEY POINT**
> Manipulation = **transforming and organising** data so it can be analysed. Pandas is the most common Python library for it.

- Filtering rows by a condition
- Selecting or removing columns
- Sorting values
- Grouping and aggregation (sum, mean, count)
- Merging / joining datasets
- Reshaping data; changing data types
- Creating calculated (derived) columns

```
import pandas as pd
df = pd.DataFrame({"city": ["Lahore", "Karachi", "Lahore"], "marks": [78, 91, 70]})
print(df.groupby("city")["marks"].mean())
```

Needs pandas installed. Output: Karachi 91.0, Lahore 74.0 (mean of 78 and 70).

Cleaning vs manipulation: cleaning fixes quality problems; manipulation reshapes data for analysis.

5 Data visualisation: choosing a chart

KEY POINT

Visualisation = showing data in graphical form (charts, graphs) to communicate patterns, trends, distributions and relationships. **Choose the chart by the question you are answering.**

Chart	Best used for	Example
Bar chart	Comparing categories	Sales by product
Line chart	Trends over time	Monthly revenue
Pie chart	Simple part-to-whole	Market share (few slices)
Histogram	Distribution of a numerical variable	Age distribution
Scatter plot	Relationship between two numbers	Study hours vs marks
Box plot	Spread and outliers	Salary distribution
Heatmap	Many values / correlations as colours	Correlation matrix

Good practice: clear title, labelled axes, a legend when needed, readable scales and sensible colours. Avoid misleading charts (for example, a cut-off axis or a pie with many slices).

اردو مثال

مہینہ وار آمدنی دکھانی ہو تو لائن چارٹ، مختلف دکانوں کا موازنہ ہو تو بار چارٹ، اور پڑھائی کے گھنٹے بمقابلہ نمبر ہو تو اسکیٹر پلاٹ۔

📝 Mid-exam style questions (Topic 9)

List important Python libraries and their purposes.

Differentiate data cleaning and data manipulation with examples.

Explain common charts and their uses.

One-Page Cheat Sheet

1. Lifecycle

- Problem → Acquire → Explore → Clean → EDA → Features → Model → Evaluate → Deploy → Monitor
- EDA first checks: shape, info, describe, null check

2. Roles

- Scientist: models + statistics (Python, R, SQL, TensorFlow)
- Analyst: reports (Excel, SQL, Tableau, Power BI)
- Engineer: pipelines (Spark, Hadoop, Airflow, Kafka)
- ML Engineer: deploy (PyTorch, TensorFlow, Docker)

3. Big Data: 5 Vs

- Volume = how much
- Velocity = how fast
- Variety = how many forms
- Veracity = how trustworthy
- Value = how useful

4. Data types

- Structured: SQL tables
- Semi-structured: JSON, XML
- Unstructured: images, text, audio, video
- Nominal: no order · Ordinal: order · Discrete: counted · Continuous: measured

5. Acquisition

- Data = raw facts and figures
- Primary: interview, survey, questionnaire
- Secondary: repository, database, cloud
- Check permission, privacy, quality, bias

6. Statistics

- Mean = sum / count · Median = middle · Mode = most frequent
- Range = max − min
- Variance = average squared deviation · SD = √variance
- z = (x − mean) / SD
- Normal: 68 / 95 / 99.7 % within 1 / 2 / 3 SD
- r = Cov(X,Y) / (sx·sy); correlation ≠ causation
- Hypothesis: H0 → H1 → α=0.05 → test → p → compare → decide

7. Classification metrics

- Accuracy = (TP+TN)/Total
- Precision = TP/(TP+FP)
- Recall = TP/(TP+FN)
- Specificity = TN/(TN+FP)
- F1 = 2PR/(P+R)
- FP = Type I · FN = Type II
- Example 40/45/5/10: acc 0.85, prec 0.889, rec 0.80, F1 0.842
- Imbalanced data: do not trust accuracy

8. Over / underfitting

- Underfit: train high error, test high error (high bias)
- Overfit: train very low, test high, wide gap (high variance)
- Good fit: both low and close
- Fix underfit: more complexity / features
- Fix overfit: simplify, regularise, more data, cross-validate

9. Tools and charts

- NumPy numbers · Pandas data · Matplotlib/Seaborn charts · scikit-learn ML
- Orange: visual, teaching · Weka: Java ML · Python: everything
- Bar categories · Line trend · Pie part-to-whole · Histogram distribution · Scatter relationship · Box spread/outliers